

Institute of Actuaries of India

Subject ST6 – Finance and Investment B

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Close form solution	Monte Carlo approach
Use black-scholes model to calculate guarantee or use replicating asset and value the assets	Run the simulation take average of scenario
Steps: Construct replicating portfolio with asset Use market information as input parameters for BS model	Steps: 1) Generate Risk neutral Scenario 2) match the volatility of the asset backing the liabilities 3) run simulation 4) Calculate the value of guarantee as average of all the scenarios
Need to identify basket option reflective of the assets, expert judgement would be needed. May lead to basis risk	Need to generate correlated random number and risk drivers, require to calibrate various risk drivers using market information, high spec system requirement, longer time to produce the results due to multiple simulation and complexity of modelling
No interaction of risk drivers are assumed explicitly, the construction of replicating asset would intrinsically allow for these interactions	Correlation matrix need to drive the interaction of various risk drivers like interest rate, corporate bond, equities etc

[7]

- ii) 40% fall in the equity market would result in policy holder profit increasing by risk free rate as per the product description,

Initial asset mix		Actual exposure	-40%
Equity	50%	600	360
Cbond	30%	360	360
Government bond	20%	240	240
property	0%	0	0
	100%	1200	960
Policy holder benefit change	% Change		4%(Risk free rate floor)

The new balance sheet would look like

Balance sheet t=0		Delta 12%
		40% fall
Assets		
Fund value	1200	960
Company Cash	600	600
Total assets	1800	1560
Liabilities		
policyholder Share	1300	1352
guarantees cost	300	328.8
Total Liability	1600	1680.8
Free capital	200	-120.8

Hence there is a need of increased capital of 120m

[7]

- iii) The company would have to retain 75% of the increase in the equity market and can sustain approx 20% fall in the market.

The strategy to be in place is using collar with combination of Writing Call

Buying Put option around 20% of current strike

Buying a call option of around 40% of the current strike (this is to reduce risk of sudden increase in the equity market which would lead to increase in policyholder payout)

Company should look to have low cost of hedge

Hedge effectiveness is the extent to which a hedge transaction results in offsetting changes in fair value or cash flow that the transaction was intended to provide (as identified by the hedging entity). For example, a hedge is considered to be highly effective if the changes in fair value or cash flow of the hedged item and the hedging derivative offset each other to a significant extent (80% to 125%)

[7]

[21 Marks]

Solution 2:

- i) Cumulative inflation impact should be floored at Zero hence the benefit cannot go below the current amount, further on benefit is capped at 5% so the overall equation would be

The value of annual payment given by the following equation

$$X = \max(A, \min(Q(n), B))$$

$Q(n)$ = accumulation of RPI index

$$B = (1.05)^{10}$$

If $Q(n)$ is less than A the victim effectively has a European put option which gives him the right to sell (at the vesting date i.e. time 10) his payout of $Q(n)$ per annum for an annuity of A per annum.

If $Q(n)$ is greater than the upper limit of $B = R(n) = 1.05^{10}$, then the member has written a European call option for the company. The company has the right to buy the victim's payout of $Q(n)$ per annum with an annuity of $R(n)$ per annum. [6]

ii) Uses Black's formula for the above two options

$$V_A = -Ae^{-\delta T} \Phi(f_1) + Q_0 e^{-\eta T} \Phi(f_2)$$

$$f_1 = \log(Ae^{-\delta T} / Q_0 e^{-\eta T}) / \sigma \sqrt{T} + \sigma \sqrt{T} / 2$$

$$f_2 = \log(Ae^{-\delta T} / Q_0 e^{-\eta T}) / \sigma \sqrt{T} - \sigma \sqrt{T} / 2$$

$$V_B = Q_0 e^{-\eta T} \Phi(d_1) - B e^{-\delta T} \Phi(d_1)$$

$$d_1 = \log(Q_0 e^{-\eta T} / B e^{-\delta T}) / \sigma \sqrt{T} + \sigma \sqrt{T} / 2$$

$$d_2 = \log(Q_0 e^{-\eta T} / B e^{-\delta T}) / \sigma \sqrt{T} - \sigma \sqrt{T} / 2$$

Value of benefit is = value of 1 unit linked to the RPI + value of the put option () + value of the call option () [6]

[12 Marks]

Solution 3:

i) As per the Merton model Equity of company can be considered as a put option on company's enterprise value with strike price of Debt hence risk in equity would effectively means that company should be able to service the debt better.

There are multiple model used for determining the debt of the company including structure model and reduced form model. [4]

ii) The following model uses relationship between equity volatility level bond default with β defined as relationship between them .

For calibrating the model equity volatilities can be extracted from the option market and long term volatility can be calculated using historical data.

Key limitation

- Increase equity market may lead to higher volatility though not necessarily higher default
- Data may not be available

- B calibration may require expert judgement.

[5]

iii) The following table calculate the default value and the 6% default is closer to the par value of the bond.

		Expected equity market		1	Default happens at middle of the year				
Coupon rate	8%	interest rate	5%						
Default rate	4%								
recovery	50%								
Nominal	-1000								1026.1924
Year Cashflow	Start	Start of year	Default During the year	recovery amount	End of year capital	Coupon	Redemption	total(Coupon+ recovery)	Discounting
1		1,000	40	20	960	80		100	0.952381
2	80	960	38	19	922	77		96	0.9070295
3	80	922	37	18	885	74	885	977	0.8638376
Default rate	6%								
Nominal	-1000								1000
Year Cashflow	Start	Start of year	Default During the year	recovery amount	End of year capital	Coupon	Redemption	total(Coupon+ recovery)	Discounting
1		1,000	60	30	940	80		110	0.952381
2	0	940	56	28	884	75		103	0.9070295
3	0	884	53	27	831	71	831	928	0.8638376

[8]

[17 Marks]

Solution 4:

- i) Apply the Black Scholes formula given by $C = SN(d_1) - X e^{-rT}N(d_2)$
Substituting the numbers, this works out to 2.374

[3]

- ii) Delta = $N(d_1)$

$$\text{Gamma} = \frac{N'(d_1)}{S\sigma\sqrt{T}}, \text{ where } N'(d_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{d_1^2}{2}}.$$

Substituting the values, we get Delta = 0.562 and Gamma = 0.0747

[4]

- iii) To ensure delta neutrality, the needs to buy shares and thus borrow money. Let us say the writer buys “n” shares and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n(50) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n(1) = 0$

a) This implies the number of shares to be bought is 562

b) The amount of money to be borrowed is this 25726

[2+2=4]

- iv) If the stock price moves to 52, the delta will have to be recalculated. The new delta for the updated price is 0.7017.

This implies that the total number of shares required to be delta neutral on this day is 701.7 or approximately 702.

Thus the writer needs to buy an addition 139.7 or approximately 140 shares to remain delta neutral.

[3]

- v) To ensure delta neutrality, the needs to buy shares, options and thus borrow money. Let us say the writer buys “n₁” shares, “n₂” options and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n_1(50) - n_2(1.1466) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n_1(1) - n_2(0.2965) = 0$

Portfolio has to be gamma neutral: $1000(0.0747) - n_2(0.0529) = 0$

a) This implies that the number of new options (option 2) that has to be bought is 1412.10 or approximately 1412.

b) This implies the number of shares to be bought is 143.31 or approximately 143

c) The amount of money to be borrowed is this 6410.75

d) The reason the amount of money required is less is because of the presence of a large number of options where the current outflow is the premium only.

[2+2+1+1=6]

[20 Marks]

Solution 5:

- i) Rearrange the AR(1) process $X_t - b = a(X_{t-1} - b) + \varepsilon_t$ to yield $X_t - X_{t-1} = (1 - a)(b - X_{t-1}) + \varepsilon_t$, which is a discrete time analog of the Vasicek model.

Further, the autocorrelation of X_t at lag one is a , which translates to $(1 - \beta dt)$ in the Vasicek model.

[2]

- ii) The spot rate curve is

$$r(t, T) = (-\ln P(t, T)/(T - t)) = -\ln A(t, T)/(T - t) + B(t, T)r(t, T)/(T - t)$$

So the spot rate volatility becomes

$$\frac{\partial r(t,T)}{\partial r} \sigma(r,t) = \frac{\sigma(r,t)B(t,T)}{T-t} \quad [3]$$

- iii) Changes in slope are due to changes in a, changes in curvature are due to changes in b and parallel moves are due to changes in r. [3]
- iv) The normal model does, for the following reason: It has fatter left tails and thinner right tails for probability distributions of interest rates in the future. It therefore gives thinner left tails and fatter right tails for the probability distribution of bond prices. [2]
- v) If every node contains forward rates up to the maturity of the underlying bond, a term structure up to that maturity can be constructed. The tree then needs to extend to only the maturity of the claim [3]

[13 Marks]

Solution 6:

- i) Let us first calculate the price of the zero coupon bonds from time point 3 to time point 6. The value of the zero coupon bond with maturity t is given as **ZCB (t) = exp(-t i_t)**. Thus, we have

Term (t)	3.0	3.5	4.0	4.5	5.0	5.5	6.0
Risk Free Rate (i _t)	1.15%	1.32%	1.48%	1.64%	1.80%	1.94%	2.08%
ZCB (t)	0.96609	0.95484	0.94252	0.92886	0.91393	0.89880	0.88267

Then the forward swap rate is

$$2 * (ZCB_3 - ZCB_6) / [(ZCB_{3.5} + ZCB_4 + ZCB_{4.5} + ZCB_5 + ZCB_{5.5} + ZCB_6)] = 3.02\% \quad [4]$$

ii)

- a) Value of a stream of payments of 0.5 at times 3.5,...,6 is
 $A = 0.5 * ZCB_{3.5} + ZCB_4 + ZCB_{4.5} + ZCB_5 + ZCB_{5.5} + ZCB_6 = 2.76081$

Value of the FRA is then Notional * A * (Fixed rate – Swap rate)

$$= 100,000,000 * 2.76081 * (3.5\% - 3.02\%) = \text{Euro } 1.321 \text{ million} \quad [2]$$

- b) This is an amount that the financial institution would pay to the counterparty. [2]

iii) Price of a receiver swaption is given by

$$LA[s_k N(-d_2) - s_0 N(-d_1)]$$

Where:

$$\bullet d_1 = \frac{\ln\left(\frac{s_0}{s_k}\right) + \frac{\sigma^2 T}{2}}{\sigma \sqrt{T}}$$

- $d_2 = d_1 - \sigma\sqrt{T}$
- σ is the Black volatility = 0.2
- s_0 is the forward swap rate = 3.02%
- s_k is the at – the – money strike = 3.02%
- L is the notional principal = Euro 100 million
- A , as in part (b), is the value of a stream of payments at times 3.5, 4.0, 4.5, 5.0, 5.5 and 6.0. This value is 2.76081
- T is 3 years

Thus, in this case

- $d_1 = 0.17321 \Rightarrow N(d_1) = 0.43125$
- $d_2 = -0.17321 \Rightarrow N(d_2) = 0.56785$

Hence, the swaption price is

$$\text{Euro } [100,000,000 * 2.76081 * 3.02\% * (0.56785 - 0.43125)] = \text{Euro } 1.147\text{million} \quad [5]$$

iv) The purchase of a FRA hedges the financial institution's market risk perfectly.

The swaption, on the other hand:

- Allows the financial institution to profit from interest rate increases: if the 3 year swap rate at time 3 exceeds 3.5%, then the firm benefits from earning a higher rate on the Euro 100million than it is paying out (with the FRA, the firm would be hedging away profits as well as losses).
- Under-protects the firm by having a strike rate of 3.02% rather than the desired 3.5%. If the three year swap rate at time $t = 3$ is below 3.02%, the firm will lose 0.48% on a net basis, this being the difference between the 3.02% received and the 3.5% paid. If the three year swap rate at time 3 is between 3.02% and 3.5%, the swaption expires worthless and the firm is left to pay out 3.5%, which exceeds the swap rate at that time.

As a result of the value of the under-hedging exceeding the value of the extra profits in the "interest rate up" scenarios, the FRA is more expensive than the swaption.

The firm needs to consider also the credit risk associated with the two possible hedges. Today the credit exposure is at a similar level for the two hedges. If interest rates increase above 3.5%, the FRA will become a liability to the financial institution (so not a credit risk) and the swaption would have a small time value.

If interest rates remain below 3.5%, then close to maturity the FRA would be worth more than the swaption so represents a bigger credit risk, and further from maturity the swaption will have some time value, so things will be less clear.

It is also worth noting that the FRA (but not the swaption) can become a liability to the financial institution and an asset to the counterparty. If the counterparty has insisted on credit risk mitigation processes being in place, the financial institution could find itself being called upon to post collateral with the counterparty.

The swaption will also expose the financial institution to the risk arising out of changing volatility; in other wordsvega risk.

[4]

[17 Marks]
